

DRUMHELLER EAST, AB, Canada

WMO: 712370

Lat: 51.450N

Lon: 112.680W

Elev: 678

StdP: 93.44

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.0	(c) -27.9	(d) -35.9	(e) 0.1	(f) -32.0	(g) -31.5	(h) 0.2	(i) -27.7	(j) 9.4	(k) -3.2	(l) 8.4	(m) -4.0	(n) 1.1	(o) 160	(p) 0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.0	(c) 31.6	(d) 17.1	(e) 29.5	(f) 16.4	(g) 27.5	(h) 15.9	(i) 19.0	(j) 27.3	(k) 17.9	(l) 26.6	(m) 17.0	(n) 24.9	(o) 3.5	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	12.5	22.1	14.9	11.5	20.8	13.8	10.7	19.8	56.8	27.2	53.2	26.7	50.1	24.9	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.5	6.5	-35.9	34.8	3.9	1.9	-38.7	36.1	-40.9	37.2	-43.1	38.2	-45.9	39.6		
			-36.0	20.8	3.8	1.4	-38.8	21.8	-41.0	22.6	-43.2	23.3	-45.9	24.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-11.5	-8.9	-2.4	5.4	11.6	15.7	18.9	17.9	12.8	5.4	-3.4	-10.3
	DBStd	12.32	8.56	7.74	7.95	5.02	4.34	3.21	3.03	3.33	4.34	4.74	7.23	7.83
	HDD10.0	2972	667	531	386	148	33	1	0	1	21	153	403	628
	HDD18.3	5212	926	764	643	387	211	91	30	49	173	400	653	886
	CDD10.0	903	0	0	1	12	83	172	276	244	104	12	1	0
	CDD18.3	102	0	0	0	0	3	12	47	35	6	0	0	0
	CDH23.3	1829	0	0	1	8	114	221	679	598	199	9	0	0
	CDH26.7	597	0	0	0	1	24	49	237	221	63	1	0	0
(14) Wind	WSAvg	2.6	2.1	2.2	2.7	3.2	3.3	3.0	2.8	2.6	2.6	2.6	2.3	2.0
(15) Precipitation	PrecAvg	384	14	10	16	26	49	77	65	54	32	16	14	12
	PrecMax	490	34	24	54	60	93	152	151	108	82	35	48	29
	PrecMin	229	1	1	3	3	6	33	24	3	5	5	2	1
	PrecStd	65	8	6	10	14	24	30	30	28	19	9	11	8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	9.7	17.5	24.2	29.6	30.7	33.7	34.2	31.9	24.4	15.9	10.4
		MCWB	3.9	4.1	7.8	11.3	14.1	17.0	18.2	17.3	15.8	12.3	7.8	4.3
	2%	DB	6.4	6.6	13.6	21.1	26.5	27.9	31.6	31.5	28.4	19.8	12.0	6.7
		MCWB	2.7	2.4	6.1	9.7	13.2	15.8	17.9	16.8	14.6	10.2	5.6	2.6
	5%	DB	3.7	4.3	10.8	17.9	23.9	25.8	29.3	29.2	25.4	16.8	8.8	3.8
		MCWB	0.8	1.0	4.5	8.0	12.0	14.7	17.4	16.3	13.7	8.7	3.9	0.7
	10%	DB	0.3	1.8	8.0	15.0	21.3	23.9	27.2	26.8	22.4	13.9	5.8	0.7
		MCWB	-1.6	-0.7	3.2	6.6	10.8	13.9	16.8	15.8	12.5	7.2	2.2	-1.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.6	8.6	11.8	15.3	18.8	21.1	19.8	16.7	12.9	8.2	5.2
		MCDB	8.1	9.1	16.1	23.1	26.2	27.3	29.3	28.5	28.3	23.0	15.3	9.6
	2%	WB	2.8	2.6	6.5	10.1	14.0	17.1	19.6	18.5	15.4	10.7	6.0	2.6
		MCDB	6.2	6.1	12.9	19.8	23.8	24.8	27.9	27.7	26.1	18.4	11.4	6.3
	5%	WB	0.9	1.1	4.9	8.6	12.8	15.9	18.4	17.5	14.2	9.2	4.2	0.7
		MCDB	3.6	4.0	10.4	17.0	21.7	23.3	26.6	26.4	23.9	15.7	8.2	3.6
	10%	WB	-1.6	-0.6	3.4	7.0	11.6	14.7	17.4	16.6	12.9	7.8	2.4	-1.3
		MCDB	0.2	1.8	7.7	14.1	19.9	21.9	25.4	24.7	21.6	13.3	5.5	0.8
(35) Mean Daily Temperature Range	MDBR		11.7	13.2	12.7	13.5	13.8	12.7	14.0	14.8	15.1	13.6	11.4	11.4
	5% DB	MCDBR	14.9	15.8	16.7	19.5	18.7	16.7	17.9	19.7	21.1	18.9	15.2	14.6
		MCWBR	11.5	11.8	9.9	10.0	8.4	7.0	7.0	7.8	9.3	10.1	9.5	10.9
	5% WB	MCDBR	14.5	15.3	16.2	17.9	16.6	14.5	15.7	16.9	20.0	17.7	14.2	14.2
		MCWBR	11.3	11.6	10.0	9.4	7.8	6.8	6.9	7.4	9.0	9.8	9.3	10.9
(40) Clear-Sky Solar Irradiance	taub		0.272	0.278	0.292	0.351	0.377	0.370	0.367	0.386	0.334	0.290	0.260	0.248
	taud		2.330	2.401	2.446	2.293	2.271	2.344	2.378	2.305	2.418	2.520	2.475	2.371
	Ebn at Noon		752	862	917	886	875	884	880	841	854	840	780	732
	Edh at Noon		62	77	89	118	126	119	113	115	92	68	53	50
(44) All-Sky Solar Radiation	RadAvg		1.10	2.13	3.50	4.75	5.74	6.16	6.58	5.32	3.75	2.27	1.21	0.87
	RadStd		0.08	0.17	0.30	0.37	0.38	0.37	0.32	0.38	0.39	0.27	0.10	0.09

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (79 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page